

Challenge (Habanero)

- Q1.** Factor the expression: $x^3 + 8$
 (A) $x^3 + 2^3$
 (B) $(x + 2)(x^2 - 2x + 4)$
 (C) $x(x^2 + 8)$
 (D) $(x + 2)(x^2 - 2x - 4)$
 (E) $x(x + 2)(x - 2)$
- Q2.** Simplify the expression: $(2x)^3 - 64$
 (A) $8x^3 - 64$
 (B) $8x^3 + 64$
 (C) $8(x^3 - 8)$
 (D) $8(x - 2)(x^2 + 2x + 4)$
 (E) $8(x + 2)(x^2 - 2x + 4)$
- Q3.** Factor the expression: $27x^3 - 64y^3$
 (A) $(3x)^3 - (4y)^3$
 (B) $(3x - 4y)(9x^2 + 12xy + 16y^2)$
 (C) $(3x + 4y)(9x^2 - 12xy + 16y^2)$
 (D) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
 (E) $(x - y)(27x^2 + 27xy + 27y^2)$
- Q4.** Simplify the expression: $8x^3 + 27$
 (A) $(2x)^3 + 3^3$
 (B) $(2x + 3)(4x^2 - 6x + 9)$
 (C) $(2x - 3)(4x^2 + 6x + 9)$
 (D) $(2x)^3 - 3^3$
 (E) $2x(x^2 + 9)$
- Q5.** An architect is designing a building with a cubic foundation. The length of each side of the foundation is increasing by 2 meters. The original volume of the foundation is $125m^3$. What is the new volume of the foundation?
- (A) $1331m^3$
 (B) $2197m^3$
 (C) $343m^3$
 (D) $729m^3$
 (E) $512m^3$
- Q6.** Factor the expression: $x^3 - 125$
 (A) $x^3 + 125$
 (B) $(x - 5)(x^2 + 5x + 25)$
 (C) $(x + 5)(x^2 - 5x + 25)$
 (D) $(x - 5)(x^2 - 5x - 25)$
 (E) $x(x^2 - 125)$
- Q7.** Simplify the expression: $(x + 2)^3 - (x - 2)^3$
 (A) $12x$
 (B) $12x^3$
 (C) $12x^2$
 (D) $12x^2 + 8$
 (E) $12x^2 - 8$
- Q8.** A recipe for a dish calls for $\frac{1}{8}$ teaspoon of a spice. If the recipe is doubled, and then the amount of spice is increased by $\frac{1}{8}$ teaspoon, what is the new amount of spice needed?
 (A) $\frac{14}{8}$ teaspoon
 (B) $\frac{12}{8}$ teaspoon
 (C) $\frac{38}{8}$ teaspoon
 (D) $\frac{12}{8}$ teaspoon + $\frac{18}{8}$ teaspoon
 (E) 1 teaspoon

Open Ended Questions (FRQ)

- Q9.** Factor the expression: $64x^3 + 125y^3$. Show all steps and simplify your answer.
- Q10.** A construction company is building a set of identical cubic modules for a building. Each module has a volume of $216m^3$. If the length of each side of the module is increased by 1 meter, what is the new volume of each module? Show all calculations.

Chef's Tips

- Provide clear instructions for each question.
- Allow students to use calculators only for checking calculations.
- Encourage students to show all work and explanations for free response questions.